
Wireless Digital Communications

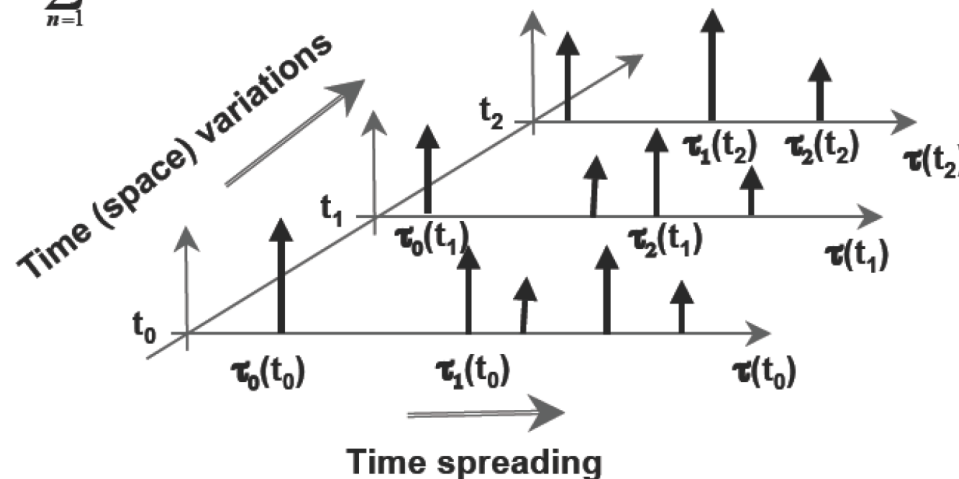
Topic 2

Capacity of Fading Channels

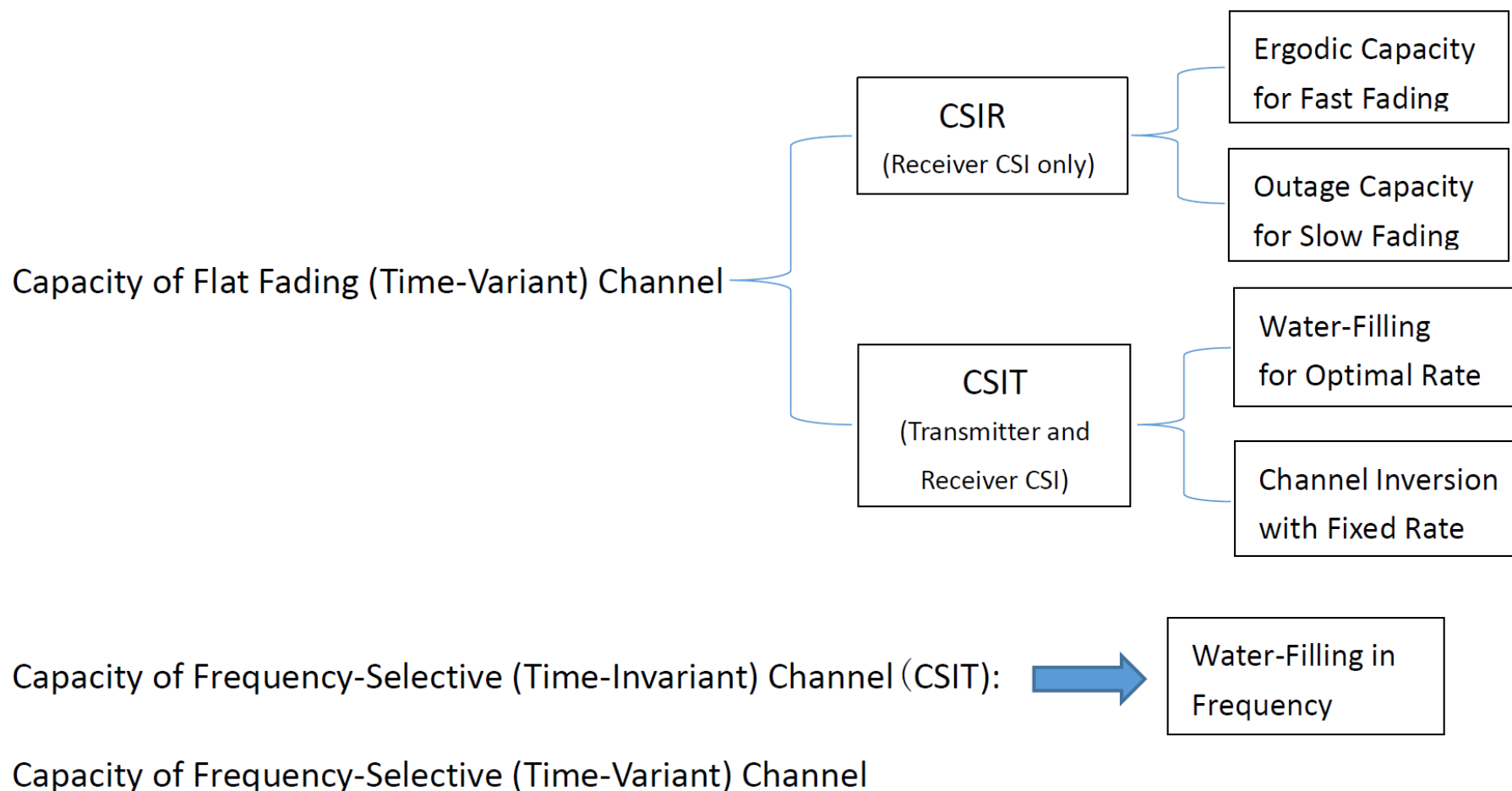
Classification of Wireless Channels

多径性	时变性
Flat Fading (<i>Narrow-Band Transmission</i>)	Time-Invariant
Frequency-Selective Fading (<i>Broad-Band Transmission</i>)	Time-Variant (<i>Fast, Slow</i>)

$$c(\tau, t) = \sum_{n=1}^N \alpha_n(t) e^{-j\varphi_n(t)} \delta(\tau - \tau_n(t))$$



Outline



Capacity of Wireless Channels

- Pioneered by Claude Shannon in 1948
- Defined as the maximum MI of channel
- Maximum error-free data rate a channel can support.
- Theoretical limit (not achievable)
- Channel characteristic
 - Not dependent on design techniques

Basic Principles



$$H(X) = -\sum_x p(x) \log[p(x)]$$

Entropy of X, Average information of input. Uncertainty in X

$$H(Y) = -\sum_y p(y) \log[p(y)]$$

Entropy of Y, Average information of output. Uncertainty in Y

$$H(X|Y) = -\sum_{x,y} p(x,y) \log p(x|y)$$

Conditional entropy, Uncertainty in X when Y is observed

Channel Capacity

Mutual Information: $I(X,Y) = H(X) - H(X|Y)$

Uncertainty removed by transmission

Channel Capacity: $C = \max_{p(x)} I(X,Y) = \max_{p(x)} [H(X) - H(X|Y)]$
(bit/symbol)

Capacity of Additive White Gaussian Noise (AWGN) Channel

$$C = B \log_2(1 + \gamma) \quad \begin{array}{l} B = \text{Channel Bandwidth} \\ \gamma = \text{Channel SNR} = P/N_0B \end{array}$$

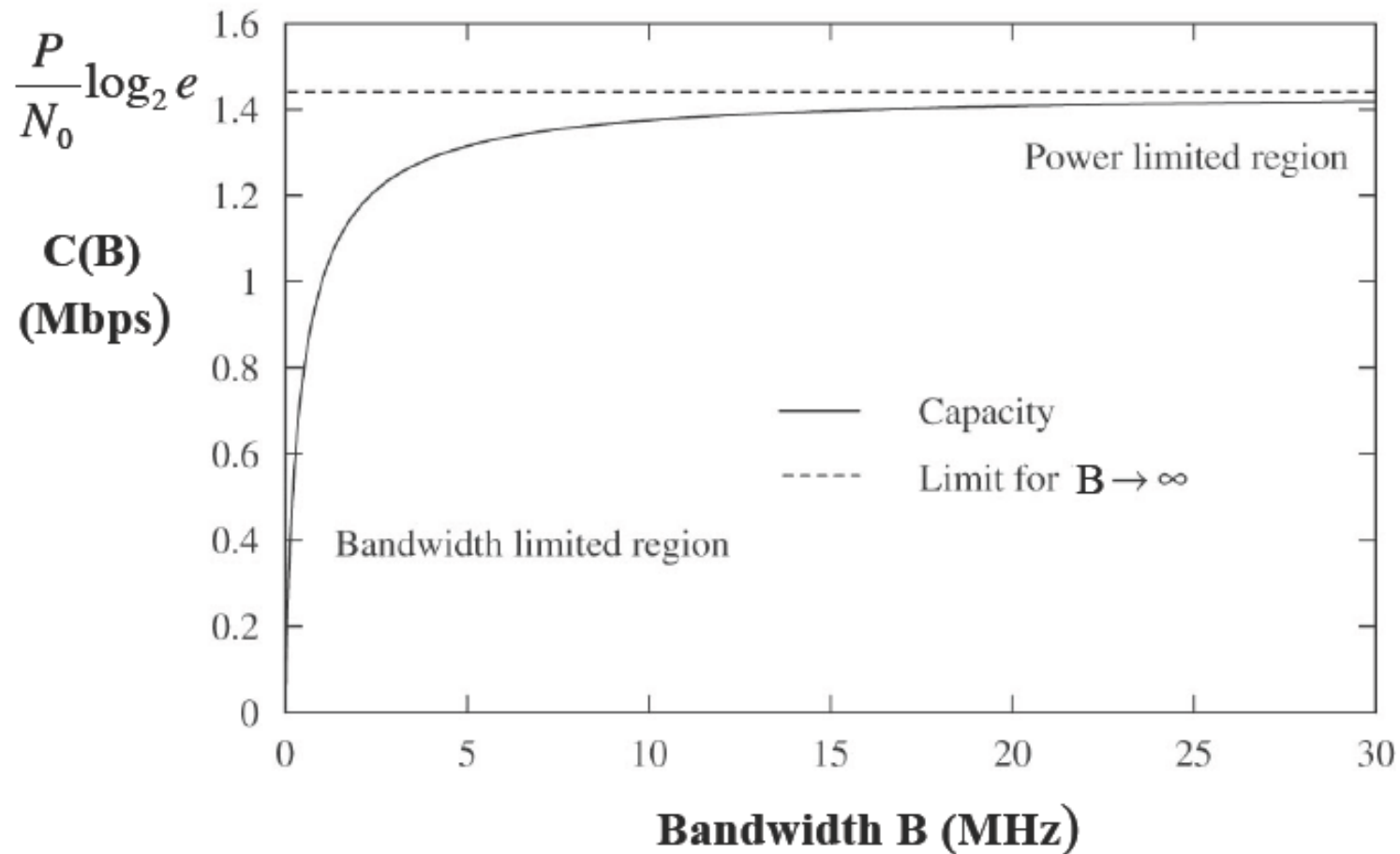
Need CSI γ at both transmitter and receiver

Example 4.1

EXAMPLE 4.1: Consider a wireless channel where power falloff with distance follows the formula $P_r(d) = P_t(d_0/d)^3$ for $d_0 = 10$ m. Assume the channel has bandwidth $B = 30$ kHz and AWGN with noise PSD $N_0/2$, where $N_0 = 10^{-9}$ W/Hz. For a transmit power of 1 W, find the capacity of this channel for a transmit–receive distance of 100 m and 1 km.

Solution: The received SNR is $\gamma = P_r(d)/N_0B = .1^3/(10^{-9} \cdot 30 \cdot 10^3) = 33 = 15$ dB for $d = 100$ m and $\gamma = .01^3/(10^{-9} \cdot 30 \cdot 10^3) = .033 = -15$ dB for $d = 1000$ m. The corresponding capacities are $C = B \log_2(1 + \gamma) = 30000 \log_2(1 + 33) = 152.6$ kbps for $d = 100$ m and $C = 30000 \log_2(1 + .033) = 1.4$ kbps for $d = 1000$ m. Note the significant decrease in capacity at greater distances due to the path-loss exponent of 3, which greatly reduces received power as distance increases.

Power and Bandwidth Limited Regions



Capacity of Flat-Fading Channels

- **Fading Value Known at Receiver Only (Receiver CSI):**
The channel gain is known to the receiver, and both the transmitter and receiver know the distribution.
- **Fading Value Known at Transmitter and Receiver (Transmitter and Receiver CSI):** The channel gain is known to the transmitter and receiver, and both the transmitter and receiver know the distribution.

Capacity of Flat-Fading Channels

Receiver CSI Only

Channel Side Information (CSI) is Channel Power Gain

Fading Value Known at Receiver Only

Fading Statistics Known at both Transmitter and Receiver

Shannon Capacity (Ergodic Capacity)

$$C = \int_0^{\infty} B \log_2(1 + \gamma) p(\gamma) d\gamma$$

$$\gamma = \frac{\bar{P}g[i]}{N_0B}$$

R. J. McEliece and W. E. Stark, "Channels with block interference," *IEEE Trans. Inform. Theory*, pp. 44–53, January 1984.

This is average capacity, lower than capacity with average γ

Capacity of Flat-Fading Channels

Receiver CSI Only

Jensen's Inequality:

$$\begin{aligned}\mathbf{E}[B\log_2(1 + \gamma)] &= \int_0^{\infty} B\log_2(1 + \gamma)p(\gamma)d\gamma \\ &\leq B\log_2(1 + \mathbf{E}[\gamma]) = B\log_2(1 + \bar{\gamma})\end{aligned}$$

Shannon capacity with receiver CSI only $<$ Channel capacity of AWGN channel with the same average γ

Fading reduces capacity with the receiver CSI only

Example 4.2

EXAMPLE 4.2: Consider a flat fading channel with i.i.d. channel gain $\sqrt{g[i]}$, which can take on three possible values: $\sqrt{g_1} = .05$ with probability $p_1 = .1$, $\sqrt{g_2} = .5$ with probability $p_2 = .5$, and $\sqrt{g_3} = 1$ with probability $p_3 = .4$. The transmit power is 10 mW, the noise power spectral density $N_0/2$ has $N_0 = 10^{-9}$ W/Hz, and the channel bandwidth is 30 kHz. Assume the receiver has knowledge of the instantaneous value of $g[i]$ but the transmitter does not. Find the Shannon capacity of this channel and compare with the capacity of an AWGN channel with the same average SNR.

Solution: The channel has three possible received SNRs: $\gamma_1 = P_t g_1 / N_0 B = .01 \cdot (.05^2) / (30000 \cdot 10^{-9}) = .8333 = -7.9$ dB, $\gamma_2 = P_t g_2 / N_0 B = .01 \cdot (.5^2) / (30000 \cdot 10^{-9}) = 83.333 = 19.2$ dB, and $\gamma_3 = P_t g_3 / N_0 B = .01 / (30000 \cdot 10^{-9}) = 333.33 = 25$ dB. The probabilities associated with each of these SNR values are $p(\gamma_1) = .1$, $p(\gamma_2) = .5$, and $p(\gamma_3) = .4$. Thus, the Shannon capacity is given by

$$\begin{aligned} C &= \sum_i B \log_2(1 + \gamma_i) p(\gamma_i) \\ &= 30000 (.1 \log_2(1.8333) + .5 \log_2(84.333) + .4 \log_2(334.33)) \\ &= 199.26 \text{ kbps.} \end{aligned}$$

The average SNR for this channel is $\bar{\gamma} = .1(.8333) + .5(83.33) + .4(333.33) = 175.08 = 22.43$ dB. The capacity of an AWGN channel with this SNR is $C = B \log_2(1 + 175.08) = 223.8$ kbps. Note that this rate is about 25 kbps larger than that of the flat fading channel with receiver CSI and the same average SNR.

Capacity of Flat-Fading Channels

Receiver CSI Only and Constant SNR over a Large Number of Transmissions

- Capacity with Outage

$$C = B \log_2 (1 + \gamma_{\min})$$

$$P_{out} = P(\gamma < \gamma_{\min})$$

- Average Rate Correctly Received

$$C_{out} = (1 - P_{out}) B \log_2 (1 + \gamma_{\min})$$

Capacity of Flat-Fading Channels

Transmitter and Receiver CSI

- Transmit power $P(\gamma)$ can be adapted
- Leads to optimization problem

$$C = \max_{P(\gamma) : E[P(\gamma)] = \bar{P}} \int_0^{\infty} B \log_2 \left(1 + \frac{\gamma P(\gamma)}{\bar{P}} \right) p(\gamma) d\gamma$$

$$\gamma = \frac{\bar{P} g[i]}{N_0 B}$$

$$\frac{P(\gamma) g[i]}{N_0 B}$$

Capacity of Flat-Fading Channels

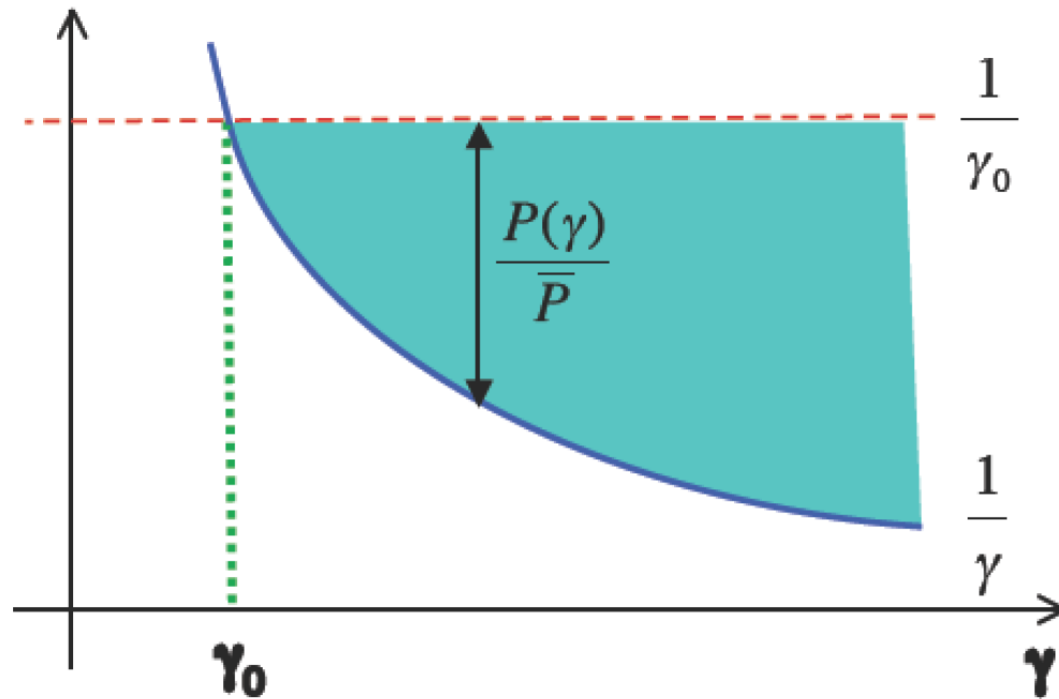
Transmitter and Receiver CSI

Optimum power allocation $P(\gamma)$ by differentiating the Lagrangian

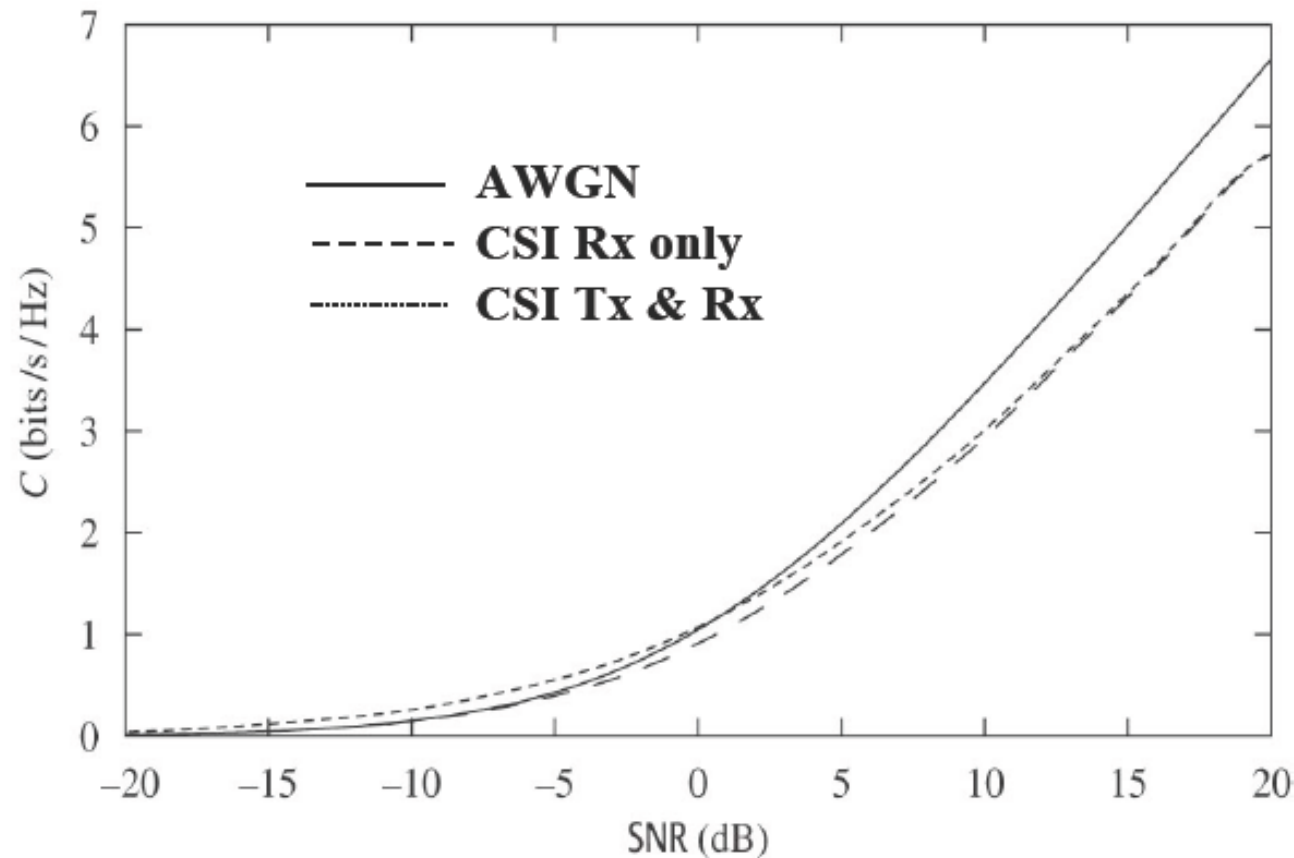
$$\frac{P(\gamma)}{\bar{P}} = \begin{cases} \frac{1}{\gamma_0} - \frac{1}{\gamma} & \gamma \geq \gamma_0 \\ 0 & \gamma < \gamma_0 \end{cases}$$
$$\int_0^{\infty} \frac{P(\gamma)}{\bar{P}} p(\gamma) d\gamma = 1$$
$$\int_{\gamma_0}^{\infty} \left(\frac{1}{\gamma_0} - \frac{1}{\gamma} \right) p(\gamma) d\gamma = 1$$
$$C = \int_{\gamma_0}^{\infty} B \log_2 \left(\frac{\gamma}{\gamma_0} \right) p(\gamma) d\gamma$$

Cutoff threshold γ depends on fading distribution $p(\gamma)$ and is obtained with power constraint formula

Optimal Power Allocation: Water-Filling Scheme

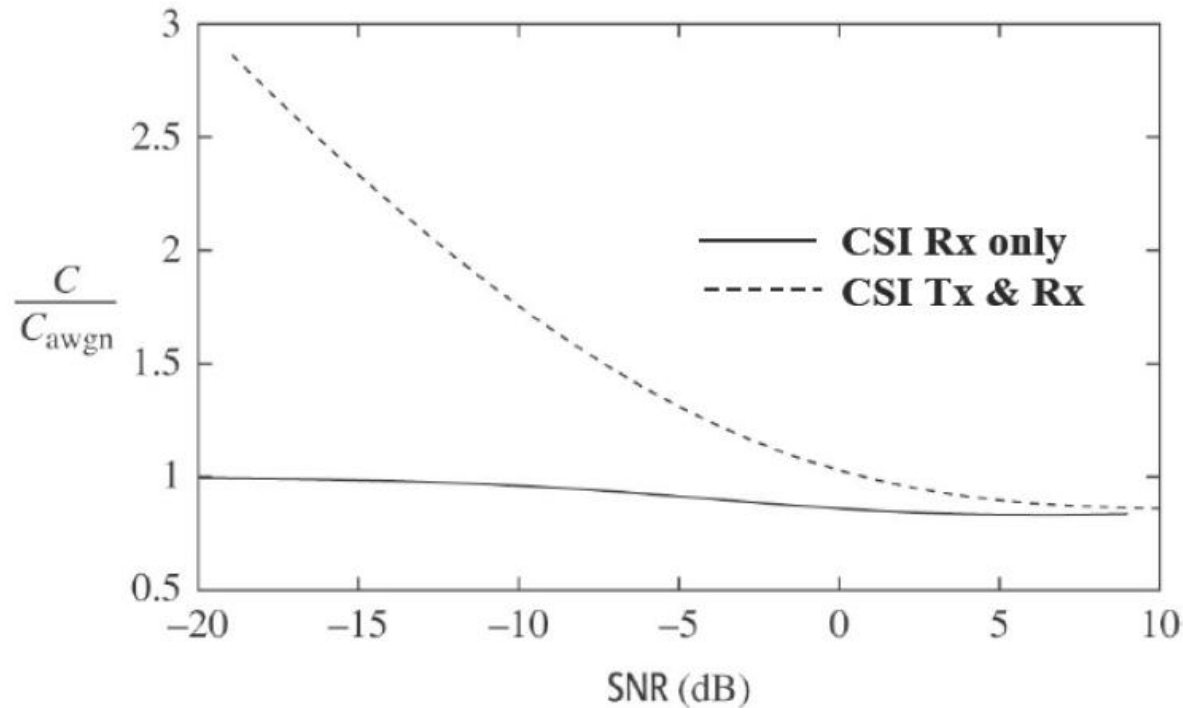


Capacity Comparisons



At high SNR water-filling does not provide much gain

Capacity Comparisons-Low SNR



Water-Filling Provides Significant Improvement at Low SNR

Example 4.4

EXAMPLE 4.4: Assume the same channel as in the previous example, with a bandwidth of 30 kHz and three possible received SNRs: $\gamma_1 = .8333$ with $p(\gamma_1) = .1$, $\gamma_2 = 83.33$ with $p(\gamma_2) = .5$, and $\gamma_3 = 333.33$ with $p(\gamma_3) = .4$. Find the ergodic capacity of this channel assuming that both transmitter and receiver have instantaneous CSI.

Solution: We know the optimal power allocation is water-filling, and we need to find the cutoff value γ_0 that satisfies the discrete version of (4.15) given by

$$\sum_{\gamma_i \geq \gamma_0} \left(\frac{1}{\gamma_0} - \frac{1}{\gamma_i} \right) p(\gamma_i) = 1. \quad (4.17)$$

We first assume that all channel states are used to obtain γ_0 (i.e., we assume $\gamma_0 \leq \min_i \gamma_i$) and see if the resulting cutoff value is below that of the weakest channel. If not then we have an inconsistency, and must redo the calculation assuming at least one of the channel states is not used. Applying (4.17) to our channel model yields

$$\sum_{i=1}^3 \frac{p(\gamma_i)}{\gamma_0} - \sum_{i=1}^3 \frac{p(\gamma_i)}{\gamma_i} = 1$$

Example 4.4

$$\Rightarrow \frac{1}{\gamma_0} = 1 + \sum_{i=1}^3 \frac{p(\gamma_i)}{\gamma_i} = 1 + \left(\frac{.1}{.8333} + \frac{.5}{83.33} + \frac{.4}{333.33} \right) = 1.13.$$

Solving for γ_0 yields $\gamma_0 = 1/1.13 = .89 > .8333 = \gamma_1$. Since this value of γ_0 is greater than the SNR in the weakest channel, this result is inconsistent because the channel should only be used for SNRs above the cutoff value. Therefore, we now redo the calculation assuming that the weakest state is not used. Then (4.17) becomes

$$\sum_{i=2}^3 \frac{p(\gamma_i)}{\gamma_0} - \sum_{i=2}^3 \frac{p(\gamma_i)}{\gamma_i} = 1$$
$$\Rightarrow \frac{.9}{\gamma_0} = 1 + \sum_{i=2}^3 \frac{p(\gamma_i)}{\gamma_i} = 1 + \left(\frac{.5}{83.33} + \frac{.4}{333.33} \right) = 1.0072.$$

Solving for γ_0 yields $\gamma_0 = .89$. Hence, by assuming that the weakest channel with SNR γ_1 is not used, we obtain a consistent value for γ_0 with $\gamma_1 < \gamma_0 \leq \gamma_2$. The capacity of the channel then becomes

Example 4.4

$$\begin{aligned} C &= \sum_{i=2}^3 B \log_2 \left(\frac{\gamma_i}{\gamma_0} \right) p(\gamma_i) \\ &= 30000 \left(.5 \log_2 \frac{83.33}{.89} + .4 \log_2 \frac{333.33}{.89} \right) = 200.82 \text{ kbps.} \end{aligned}$$

Comparing with the results of Example 4.3 we see that this rate is only slightly higher than for the case of receiver CSI only, and it is still significantly below that of an AWGN channel with the same average SNR. This is because the average SNR for the channel in this example is relatively high: for low-SNR channels, capacity with flat fading can exceed that of the AWGN channel with the same average SNR if we take advantage of the rare times when the fading channel is in a very good state.

Channel Inversion

- Fading inverted to maintain constant SNR
- Simplifies design (fixed rate)
- Greatly reduces capacity
 - Capacity is zero in Rayleigh fading
- Truncated inversion
 - Invert channel above cutoff fade depth
 - Constant SNR (fixed rate) above cutoff
 - Cutoff greatly increases capacity
 - Close to optimal

$$\frac{\gamma P(\gamma)}{\bar{P}} = \sigma$$

Channel Inversion in More Details

- Channel inversion

If target SNR is σ , transmit at $P(\gamma) = \bar{P}\sigma / \gamma$

Expected power constraint gives $\sigma = 1 / E[1/\gamma]$

What if $E[1/\gamma] = \infty$?

- Truncated channel inversion

Do not use the channel if $\gamma < \gamma_1$ (outage)

Expected power constraint gives $\sigma = 1 / E_{\gamma_1}[1/\gamma]$

Example 4.5

EXAMPLE 4.5: Assume the same channel as in the previous example, with a bandwidth of 30 kHz and three possible received SNRs: $\gamma_1 = .8333$ with $p(\gamma_1) = .1$, $\gamma_2 = 83.33$ with $p(\gamma_2) = .5$, and $\gamma_3 = 333.33$ with $p(\gamma_3) = .4$. Assuming transmitter and receiver CSI, find the zero-outage capacity of this channel.

Solution: The zero-outage capacity is $C = B \log_2[1 + \sigma]$, where $\sigma = 1/\mathbf{E}[1/\gamma]$. Since

$$\mathbf{E}\left[\frac{1}{\gamma}\right] = \frac{.1}{.8333} + \frac{.5}{83.33} + \frac{.4}{333.33} = .1272,$$

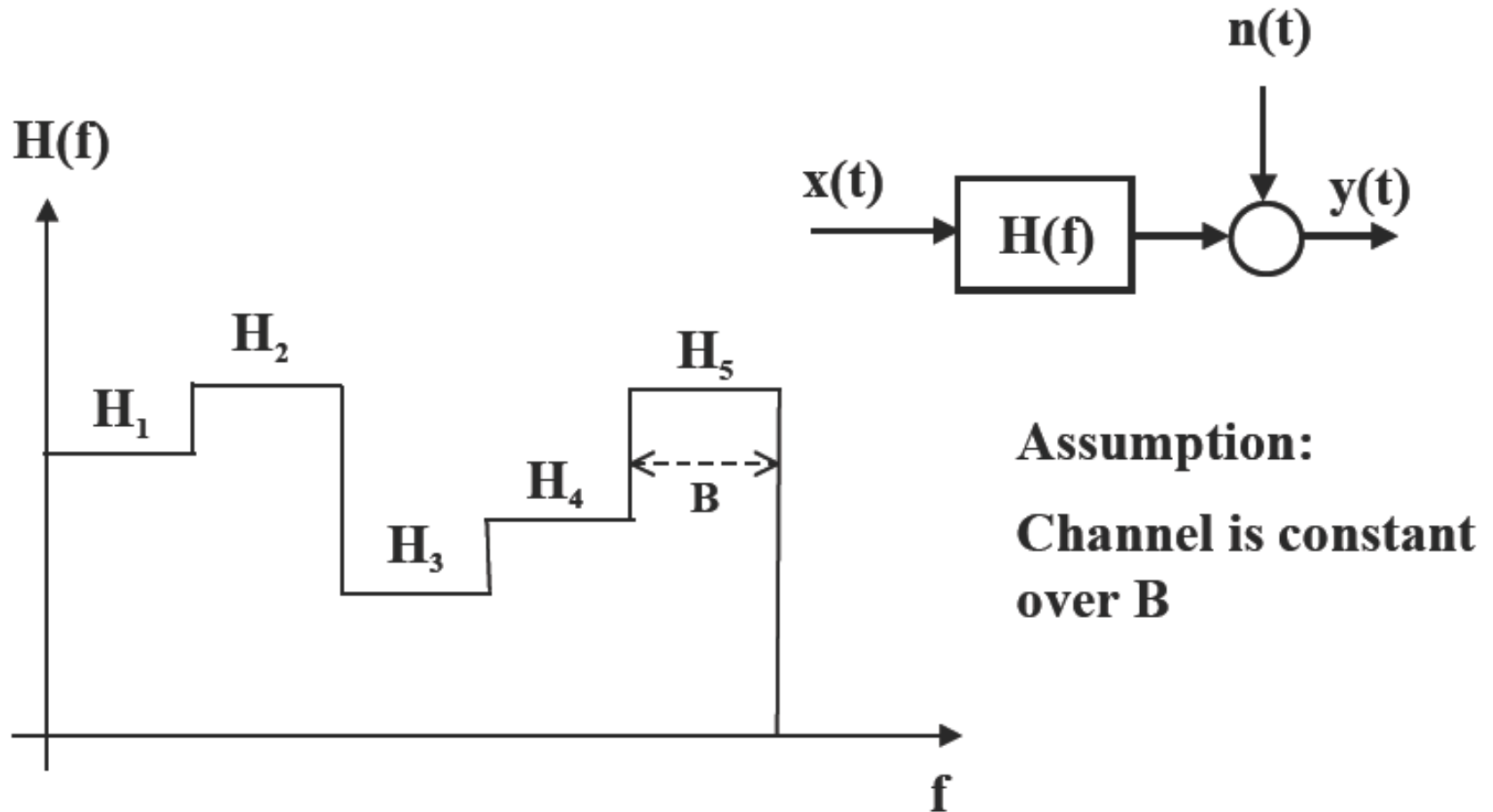
we have $C = 30000 \log_2[1 + 1/.1272] = 94.43$ kbps. Note that this is less than half of the Shannon capacity with optimal water-filling adaptation.

Example 4.6

EXAMPLE 4.6: Assume the same channel as in the previous example, with a bandwidth of 30 kHz and three possible received SNRs: $\gamma_1 = .8333$ with $p(\gamma_1) = .1$, $\gamma_2 = 83.33$ with $p(\gamma_2) = .5$, and $\gamma_3 = 333.33$ with $p(\gamma_3) = .4$. Find the outage capacity of this channel and associated outage probabilities for cutoff values $\gamma_0 = .84$ and $\gamma_0 = 83.4$. Which of these cutoff values yields a larger outage capacity?

Solution: For $\gamma_0 = .84$ we use the channel when the SNR is γ_2 or γ_3 , so $\mathbf{E}_{\gamma_0}[1/\gamma] = \sum_{i=2}^3 p(\gamma_i)/\gamma_i = .5/83.33 + .4/333.33 = .0072$. The outage capacity is $C = B \log_2(1 + 1/\mathbf{E}_{\gamma_0}[1/\gamma])p(\gamma \geq \gamma_0) = 30000 \log_2(1 + 138.88) \cdot .9 = 192.457$ kbps. For $\gamma_0 = 83.34$ we use the channel when the SNR is γ_3 only, so $\mathbf{E}_{\gamma_0}[1/\gamma] = p(\gamma_3)/\gamma_3 = .4/333.33 = .0012$. The capacity is $C = B \log_2(1 + 1/\mathbf{E}_{\gamma_0}[1/\gamma])p(\gamma \geq \gamma_0) = 30000 \log_2(1 + 833.33) \cdot .4 = 116.45$ kbps. The outage capacity is larger when the channel is used for SNRs γ_2 and γ_3 . Even though the SNR γ_3 is significantly larger than γ_2 , the fact that this larger SNR occurs only 40% of the time makes it inefficient to only use the channel in this best state.

Capacity of Frequency Selective Fading Channels (Time-Invariant)



Capacity of Frequency Selective Fading Channels (Time-Invariant)

- Frequency selective fading channel is assumed to be a set of AWGN channels in parallel, power of jth channel is P_j

- SNR of jth channel:

$$\frac{|H_j|^2 P_j}{N_0 B} \quad \gamma_j = \frac{|H_j|^2 P}{N_0 B}$$

- Capacity

$$C = \max_{P_j: \sum_j P_j \leq P} \sum_j B \log_2 \left(1 + \frac{P_j}{P} \gamma_j \right)$$

- Optimal power allocation, obtained through Lagrangian Technique leads to water-filling power allocation in frequency

Capacity of Frequency Selective Fading Channels (Time-Invariant)

$$\frac{P_j}{P} = \begin{cases} \frac{1}{\gamma_0} - \frac{1}{\gamma_j} & \gamma_j \geq \gamma_0 \\ 0 & \gamma_j < \gamma_0 \end{cases}$$

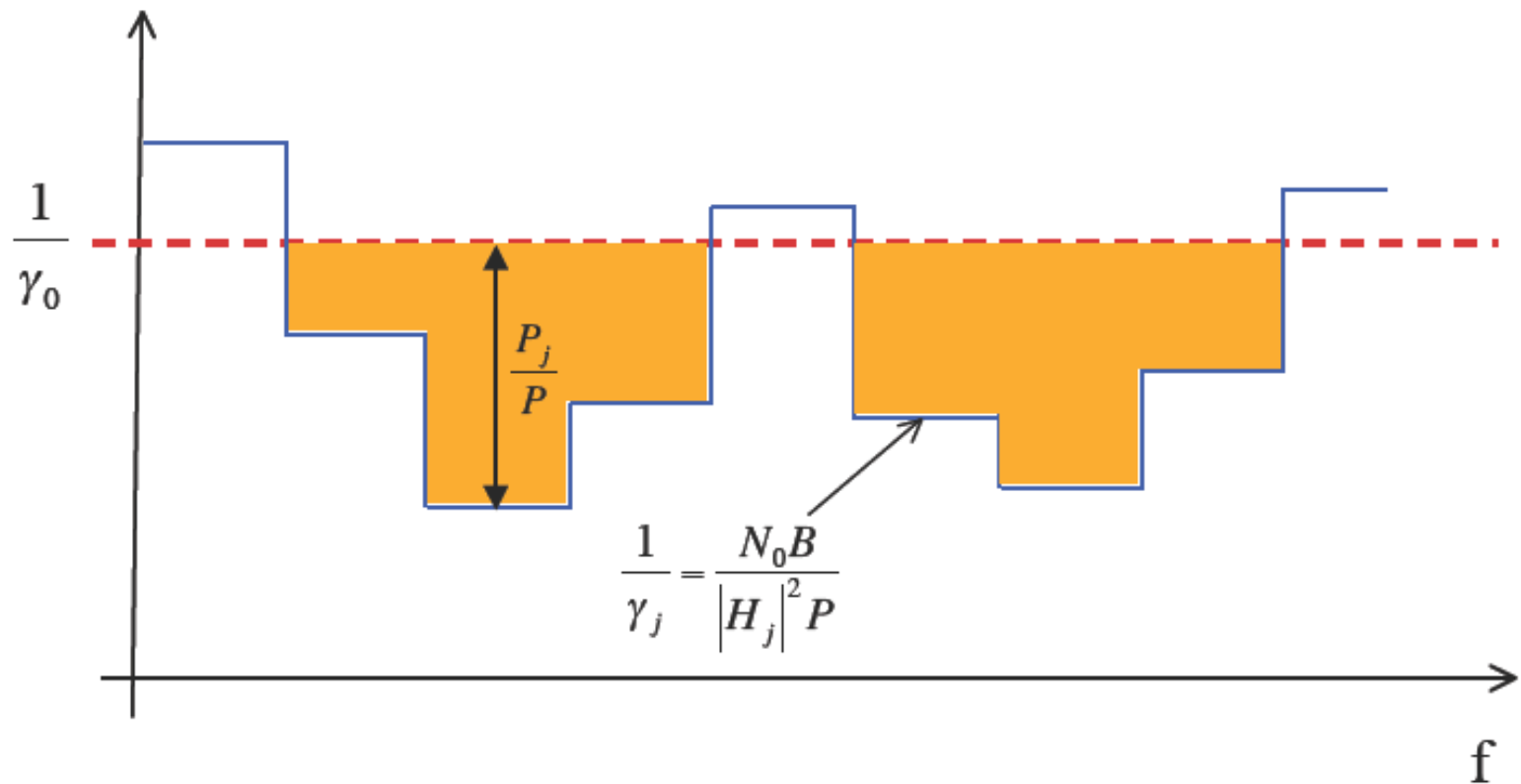
γ_0 is obtained by power constraint formula:

$$\sum_j P_j \leq P \quad \sum_j \frac{P_j}{P} = 1 \quad \sum_j \left(\frac{1}{\gamma_0} - \frac{1}{\gamma_j} \right) = 1$$

$$C = \sum_{j: \gamma_j \geq \gamma_0} B \log_2 \left(\frac{\gamma_j}{\gamma_0} \right)$$

Capacity achieved when transmitting at different rates and powers in each subchannel

Water Filling in Frequency Domain



Example 4.7

EXAMPLE 4.7: Consider a time-invariant frequency-selective block fading channel that has three subchannels of bandwidth $B = 1$ MHz. The frequency responses associated with each subchannel are $H_1 = 1$, $H_2 = 2$, and $H_3 = 3$, respectively. The transmit power constraint is $P = 10$ mW and the noise PSD $N_0/2$ has $N_0 = 10^{-9}$ W/Hz. Find the Shannon capacity of this channel and the optimal power allocation that achieves this capacity.

Solution: We first find $\gamma_j = |H_j|^2 P / N_0 B$ for each subchannel, yielding $\gamma_1 = 10$, $\gamma_2 = 40$, and $\gamma_3 = 90$. The cutoff γ_0 must satisfy (4.25). Assuming that all subchannels are allocated power, this yields

$$\frac{3}{\gamma_0} = 1 + \sum_j \frac{1}{\gamma_j} = 1.14 \implies \gamma_0 = 2.64 < \gamma_j \quad \forall j.$$

Since the cutoff γ_0 is less than γ_j for all j , our assumption that all subchannels are allocated power is consistent, so this is the correct cutoff value. The corresponding capacity is $C = \sum_{j=1}^3 B \log_2(\gamma_j/\gamma_0) = 1000000(\log_2(10/2.64) + \log_2(40/2.64) + \log_2(90/2.64)) = 10.93$ Mbps.

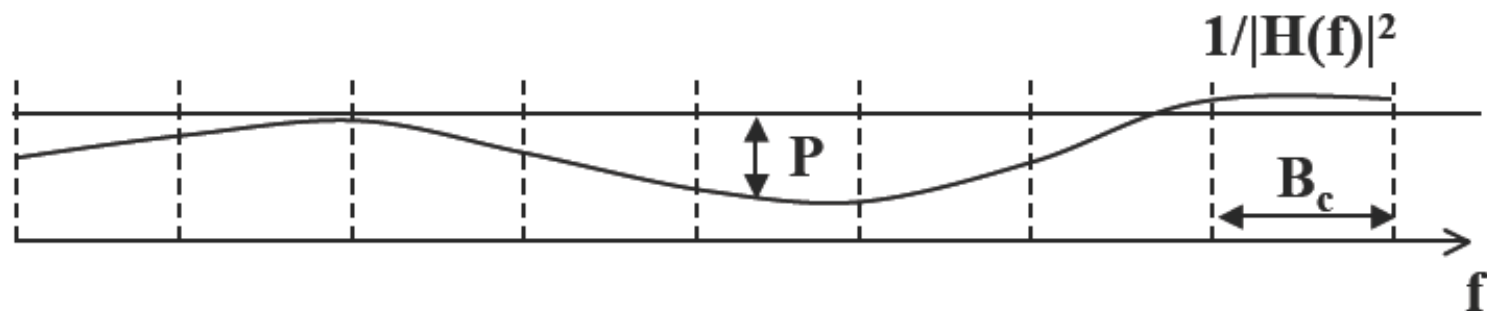
Capacity of Frequency Selective Fading Channels-Continuous Fading

Approximate by dividing into subbands

Each subband has width B_c

Independent fading in each subband

Capacity is the sum of subband capacities



$$C = \max_{P(f) : \int P(f) df \leq P} \int \log_2 \left(1 + \frac{|H(f)|^2 P(f)}{N_0} \right) df$$

Capacity of Frequency Selective Fading Channels-Continuous Fading

Using Lagrangian technique:

$$\frac{P(f)}{P} = \begin{cases} \frac{1}{\gamma_0} - \frac{1}{\gamma(f)} & \gamma(f) \geq \gamma_0 \\ 0 & \gamma(f) < \gamma_0 \end{cases}$$

where $\gamma(f) = \frac{|H(f)|^2 P}{N_0}$

$$C = \int_{f: \gamma(f) \geq \gamma_0} \log_2 \left(\frac{\gamma(f)}{\gamma_0} \right) df$$

is the resulting capacity

Main Points

- Capacity depends on degree of channel knowledge , CSI at receiver only, CSI at both receiver and transmitter
- Capacity with Transceiver CSI knowledge: variable-rate variable-power transmission (water filling) optimal
 - ▲ (Time-variant) Flat fading channels water filling in instantaneous SNR
 - ▲ (Time-invariant) Frequency selective fading water filling in frequency
- Channel inversion practical, but should truncate
- Water-filling provides significant improvement at low SNR

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- Ex. 4.5(a,b), 4.10, 4.11